MAKING RESEARCH DATA MORE ACCESSIBLE

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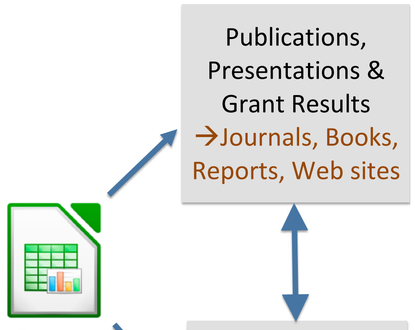
What is the Dataverse Network Project?

Science in the Digital Age

Raw Data
(collect, store & analyze)
→ Research computing facilities and domain specific databases



Publications, Presentations & Grant Results
→Journals, Books, Reports, Web sites

Final Data, Code & other Materials (Share, Cite & Archive)
→**The Dataverse Network**

The Dataverse Network is a web application for:

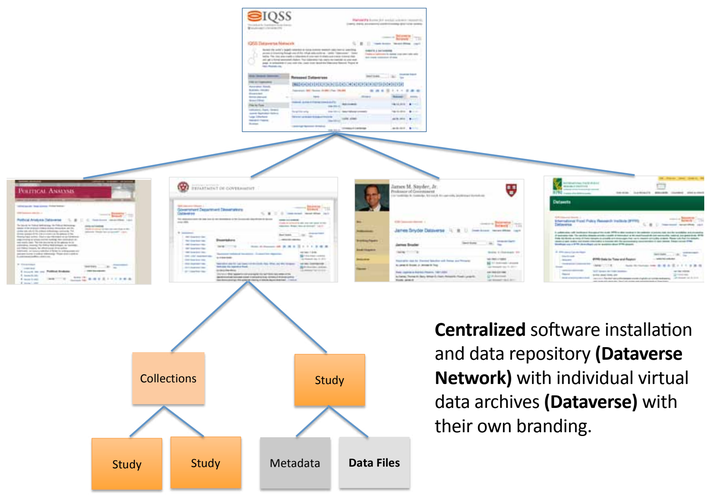
open source initiative

→Research Data

Key Benefits:

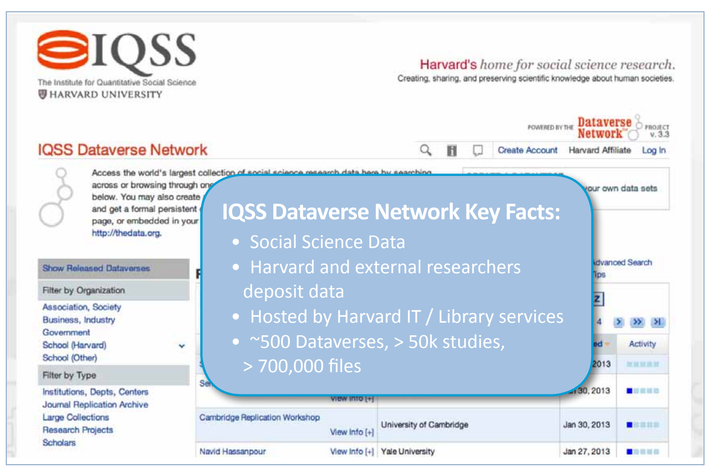
- Recognition, credit and control for data author
- Persistent citation between scholarly claims and data
- Central repository for archival and long-term access
- Validation of science through replication

Diagram of the Dataverse Network



Centralized software installation and data repository **(Dataverse Network)** with individual virtual data archives **(Dataverse)** with their own branding.

Social Science Dataverse Network

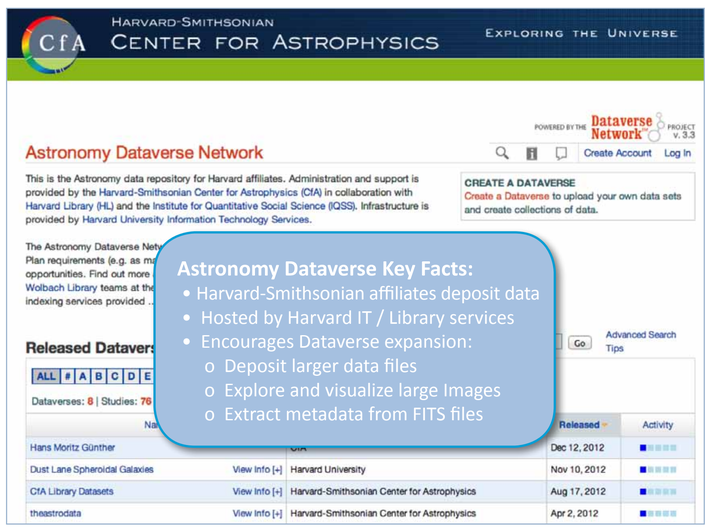


IQSS Dataverse Network Key Facts:

- Social Science Data
- Harvard and external researchers deposit data
- Hosted by Harvard IT / Library services
- ~500 Dataverses, > 50k studies, > 700,000 files

Astronomy Dataverse Network

Harvard-Smithsonian Center for Astrophysics (CfA) Astronomy Dataverse Network

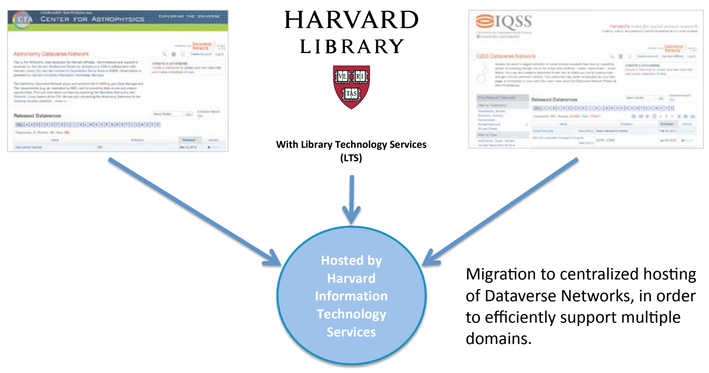


Astronomy Dataverse Key Facts:

- Harvard-Smithsonian affiliates deposit data
- Hosted by Harvard IT / Library services
- Encourages Dataverse expansion:
  - Deposit larger data files
  - Explore and visualize large Images
  - Extract metadata from FITS files

Upcoming full support for FITS astronomy data files

Harvard Library meets the Dataverse Networks



Migration to centralized hosting of Dataverse Networks, in order to efficiently support multiple domains.

Collaboration Key Facts:

- The Library/DVN project is the main solution offered at Harvard for publishing, sharing and citing research data.
- The Digital Scholarship Program at the Harvard Library, together with IQSS and research computing at Harvard, is developing other strategic projects aligned with the program's goal to enable researchers with the necessary tools to manage research data. These projects will include:
  - Providing joint resources on the research data management cycle.
  - Continuing education of librarians and researchers in data management.
  - Joint advisory services to researchers about: storage, archiving, preservation and curation of research data.

Upcoming DVN integration with Harvard Library systems:

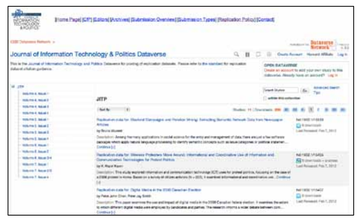
- DASH (Harvard's IR): relate publications to research data
- Digital Repository Service (DRS): as a preservation backend
- DMP (when available)

OJS-DVN Project: Developing a Data Sharing System for Journals

Funded by an Alfred P. Sloan Foundation grant, the Dataverse Network Project in collaboration with the Public Knowledge Project (PKP) are on a two-year endeavor to make data sharing and preservation an intrinsic part of the publication process.

Journal Management And Publishing System

Author submits article + research data to a journal using PKP's Open Journal System (OJS).



Journal's Research Data Repository

Research data + metadata will automatically be deposited into a journal's Dataverse repository using API technology (SWORD). In conjunction, PKP is developing a plugin to allow for this seamless integration.

Projected impacts the OJS-DVN project

"The immediate impact of the project will be to increase the number of readily replicable articles published, and the number of social science journals that adopt best data management and citation practices. The broadest impacts of the project will be to increase the pace of discovery in the social sciences, and to broaden the research opportunities for younger scholars."
- Gary King, IQSS Director and PI of Project

"This project will help establish a formal linkage between scholarly publications and the underlying research data, thus increasing replicability and reusability of previous studies to verify and advance science."
- Dr. Mercè Crosas, IQSS Director of Product Development, and Co-PI of project.

Up next for The Dataverse Network

- Expansion of the IQSS DVN to Harvard's DVN to support other domains (medical, health, etc).
- DOI support, in addition to the current Handle support for persistent identifiers.
- Integration with Shibboleth for user authentication.
- SWORD 2 compliant API to deposit research data seamlessly from Journals.
- Full support for:
  - FITS astronomy data files
  - R files
- Integration with WorldMap to visualize geospatial data.
- Support for archiving and sharing sensitive data.
- New G UI for statistical analysis and subsetting of quantitative data sets, using Zelig statistical package.